

DIGITAL IMAGE PROCESSING

LECTURE # 13

MORPHOLOGICAL IMAGE PROCESSING-I

INTRODUCTION

- The word Morphology denotes a branch of biology that deals with the form and structure of animals and plants.
- Here we use the same word in the context of Mathematical Morphology, which means as a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons etc.
- The language of Mathematical Morphology is set theory.
- Sets in Mathematical Morphology represents objects in an image.
- We do Morphology for shape analysis & shape study.
- Shape analysis became easy in case of binary images.

Basic Concepts From Set theory

Let A and B be sets.

To indicate 'a' is an element of A write $a \in A$.

To indicate a is not an element of A write $a \notin A$.

A is a subset of B , written $A \subseteq B$, if for every $a \in A$, $a \in B$.

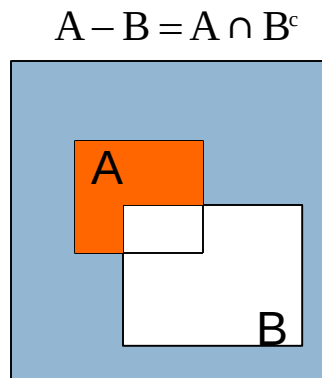
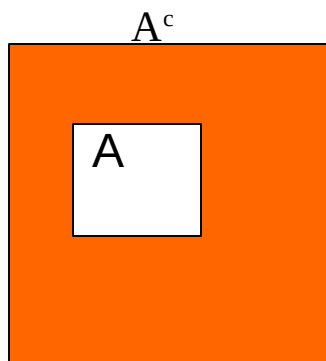
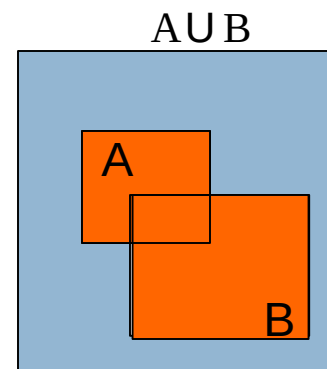
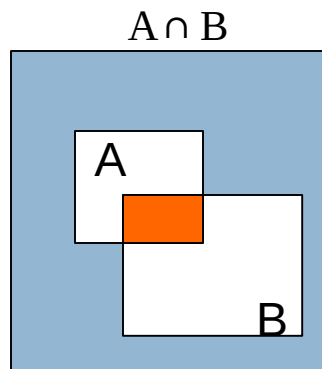
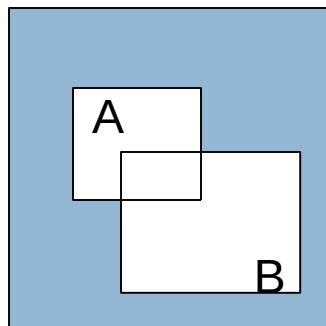
The union of A and B , $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.

The intersection of A and B , $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.

The complement of A , $A^c = \{x \mid x \notin A\}$.

The difference of A and B , $A - B = \{x \mid x \in A \text{ and } x \notin B\}$.

Examples



Basic Concepts From Set theory

- ❑ In binary images, the sets are members of the 2D integer space $\mathbb{Z}^2$, where each element of a set is a tuple (2D vector) whose coordinates are the (x, y) coordinates of a white (or black) pixel in the image.
- ❑ Gray-scale images can be represented as sets, whose components are in $\mathbb{Z}^3$: two components are coordinates of a pixel, and the third – its discrete intensity value

Reflection

Let A be a subset of Z^2 .

If A is a set of pixels representing an object in an image, then its reflection is the set of points in B , whose (x, y) coordinates have been replaced by $(-x, -y)$ as shown below

The reflection of A , written $\hat{A}$, is given by

$$\hat{A} = \{x \mid x = -a, \text{ where } a \in A\}.$$

$$\text{Here, } -a = -(a_1, a_2) = (-a_1, -a_2) \in A \subseteq Z^2.$$

Translation

Let A be a subset of $\mathbb{Z}^2$.

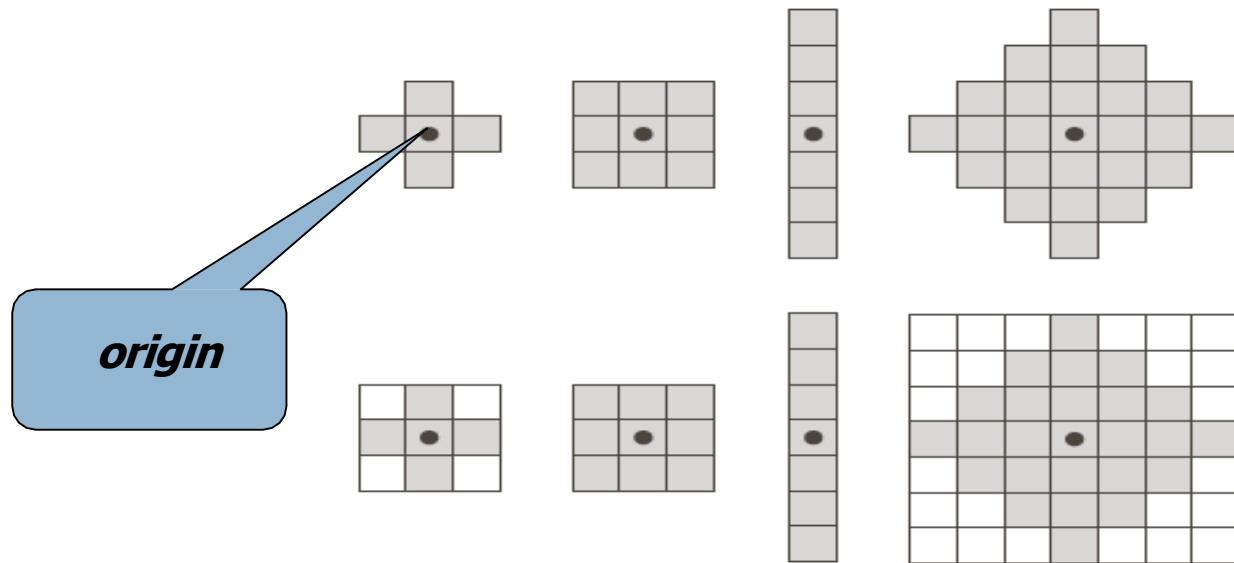
If A is the set of pixels, $(A)_z$ is the set of points, whose coordinates (x, y) were replaced by $(x + z_1, y + z_2)$.

The translation of A by z , written $(A)_z$, for some $z \in \mathbb{Z}^2$ is

$$(A)_z = \{x \mid x = a + z, \text{ for } a \in A\}.$$

Structuring Elements

- small set to probe the image under study
- for each structuring element (SE), define origin
- shape and size must be adapted to geometric properties for the objects



Fitting and Hitting

- When we place a structuring element in a binary image, each of its pixels is associated with the corresponding pixel of the neighborhood under the structuring element. In this sense, a morphological operation resembles a “binary correlation”. The operation is logical rather than arithmetic in nature.
- The structuring element is said to fit the image if, for each of its pixels that is set to 1, the corresponding image pixel is also 1.
- The structuring element is said to hit, an image if for any of its pixels that is set to 1, the corresponding image pixel is also 1.

Dilation

■ Brief Description

- ⇒ To expand or to increase something.
- ⇒ Basic effect
 - Gradually enlarge the boundaries of regions of foreground pixels on a binary image.



Common names: Dilate, Grow, Expand

Dilation

■ How It Works

A & B are sets in $\mathbb{Z}^2$, the dilation of A by B, denoted as **$A \oplus B$**

is defined as, $A \oplus B = \{z \mid (B^\wedge)_z \cap A \neq \emptyset\}$

This equation is based on obtaining the reflection of B about its origin and shifting this reflection by z.

Dilation is the set of all displacements z such that B and A overlap by at least one element

Based on this interpretation, dilation can also be represented as:

Dilation

■ Another Representation

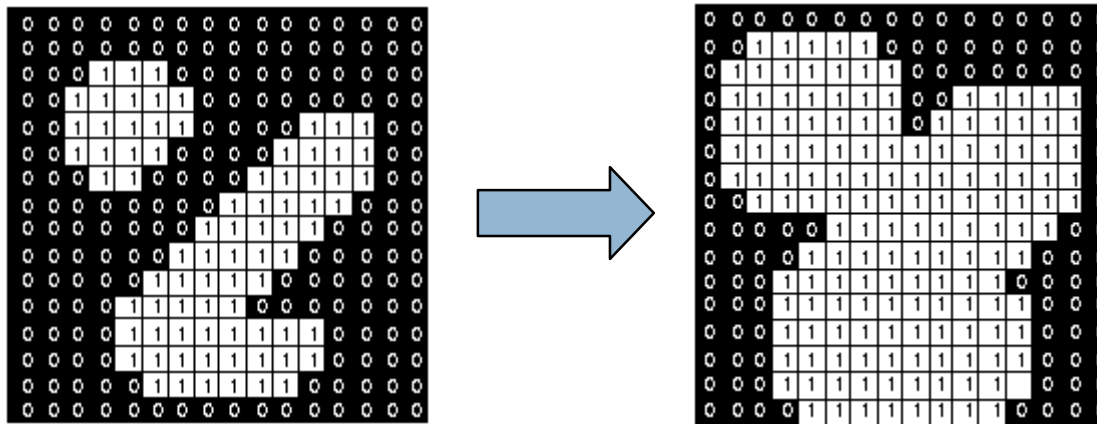
$$A \oplus B = \{z \mid (B^\wedge)_z \cap A \subseteq A\}$$

■ Does the structuring element hit the set?

Dilation of a set A by structuring element B: all z in A such that B hits A when origin of B=z

Dilation

■ Guideline for Use



Effect of dilation using a 3×3 square structuring element

1	1	1
1	1	1
1	1	1

Set of coordinate points =
 $\{ (-1, -1), (0, -1), (1, -1),$
 $(-1, 0), (0, 0), (1, 0),$
 $(-1, 1), (0, 1), (1, 1) \}$

3×3 square structuring element

Dilation Example



Erosion

■ Brief Description

- ⇒ Erosion is one of the basic operators in the area of mathematical morphology.
- ⇒ To delete or to reduce something
- ⇒ Pixels matching a given pattern are deleted from the image.
- ⇒ Basic effect
 - Erode away the boundaries of regions of foreground pixels (*i.e.* white pixels, typically).



Common names: Erode, Shrink, Reduce

Erosion

■ Operation

- ⇒ create a new set by running B over A such that the origin of B visits every element of A; at each location of the origin of B, if B is completely contained in A, mark that location as a member of a new set.
- ⇒ As a result, the boundary of A is eroded.



Common names: Erode, Shrink, Reduce

Erosion

■ How It Works

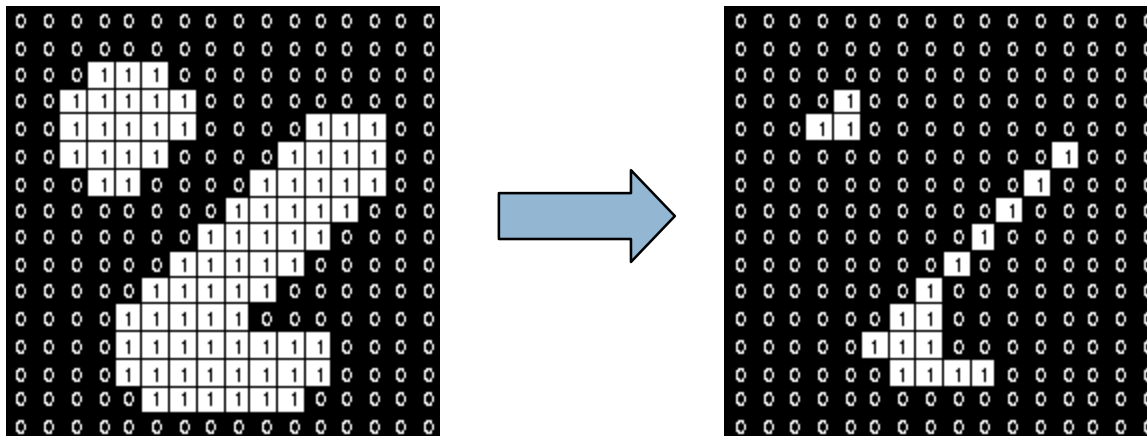
A & B are sets in $\mathbb{Z}^2$, the erosion of A by B, denoted as $A \ominus B$

$$A \ominus B = \left\{ z \mid (B)_z \subseteq A \right\}$$

- The erosion of A by B is the set of all points z such that B, translated by z, is contained in A. We will assume that the set B is a structuring element

Erosion

■ Guideline for Use



Effect of erosion using a 3×3 square structuring element

1	1	1
1	1	1
1	1	1

Set of coordinate points =
 $\{ (-1, -1), (0, -1), (1, -1),$
 $(-1, 0), (0, 0), (1, 0),$
 $(-1, 1), (0, 1), (1, 1) \}$

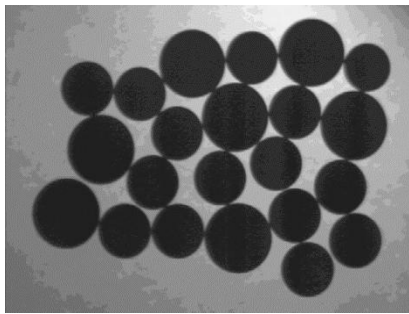
A 3×3 square structuring element

Erosion Example

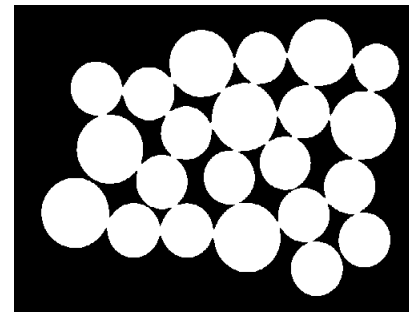


Erosion

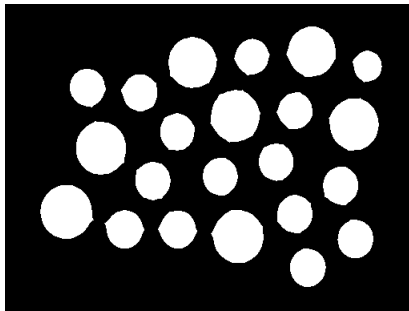
⇒ Example: Binary erosion (*separate touching disks*)



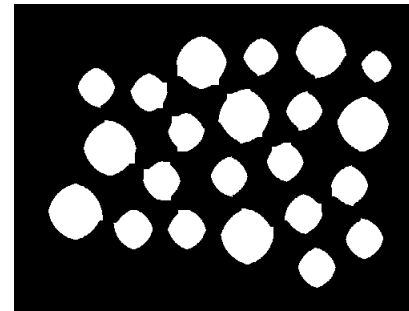
Original image (a number of dark disks)



Inverted image after thresholding



The result of eroding twice using *a disk shaped structuring element 11 pixels in diameter*



Using *9x9 square structuring element* leads to distortion of the shapes

Opening

■ Brief Description

- ⇒ The application of an erosion followed by the dilation using the same structuring element is referred to as an opening operation.
- ⇒ Opening tends to “open” small gaps or spaces between touching objects in an image.
- ⇒ Opening is also used to remove noise.
 - The erosion step in an opening will remove isolated pixels as well as boundaries of object and the dilation step will restore most of the boundary pixels without restoring the noise.



Common names: Opening

Opening

■ How It works

⇒ Opening is defined as an erosion followed by a dilation.

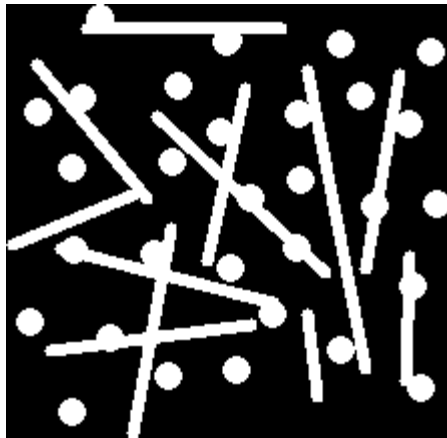
$$A \circ B = (A \ominus B) \oplus B$$

- Opening is the dual of closing.
- Opening the foreground pixels with a particular structuring element is equivalent to closing the background pixels with the same element.

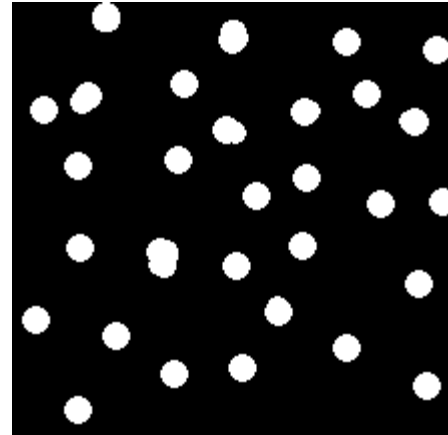
Opening

■ Binary Opening Example

⇒ Separate out the circles from the lines



A mixture of circle and lines



Opening with a disk shaped structuring element with 11 pixels in diameter

- The lines have been almost completely removed while the circles remain almost completely unaffected.

Closing

■ Brief Description

- ⇒ Closing is similar to an opening except that the dilation is performed first followed by the erosion using the same structuring element.
- ⇒ If an opening creates small gaps in an image, a closing will fill them, or “close” the gaps.
- ⇒ Closing is also used to remove noise.



Common names : Closing

Closing

■ How It works

⇒ Closing is defined as a dilation followed by an erosion.

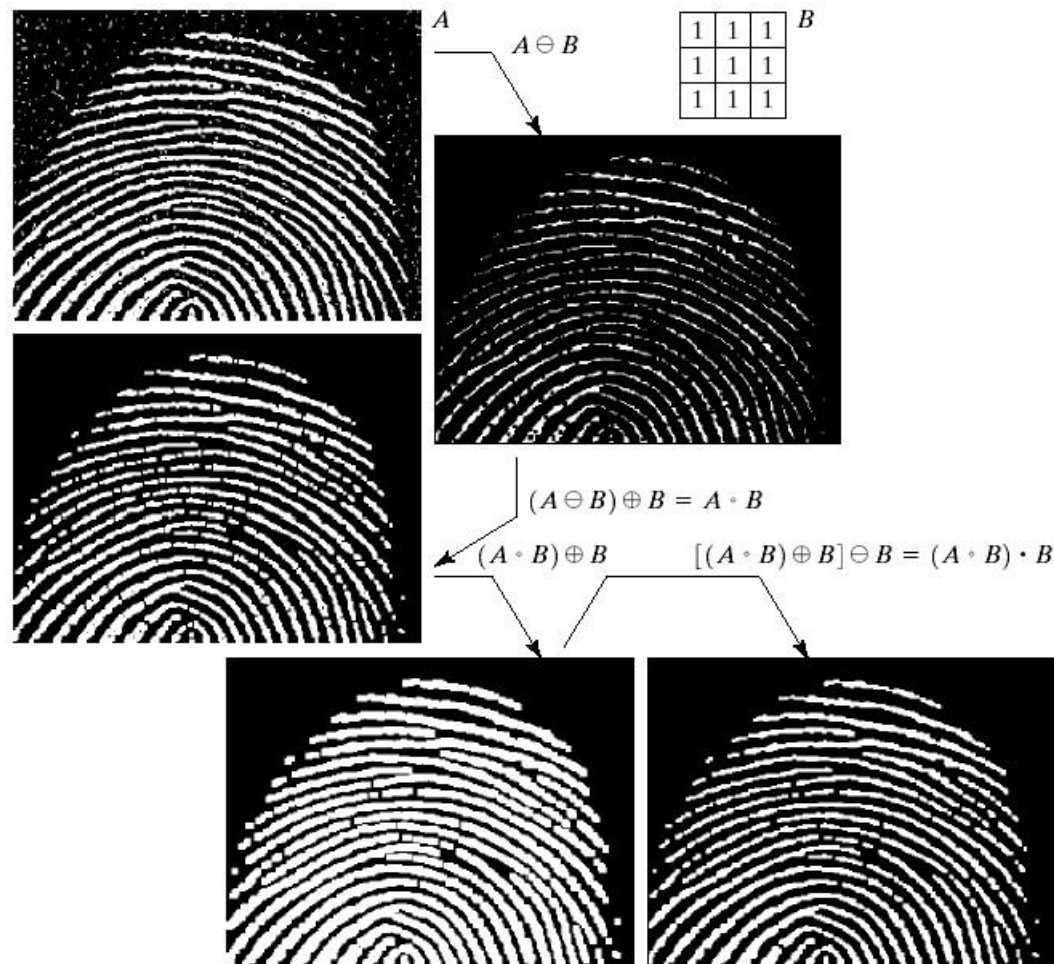
$$A \bullet B = (A \oplus B) \ominus B$$

⇒ Closing is the dual of opening.

- Closing the foreground pixels with a particular structuring element is equivalent to opening the background pixels with the same element.

Application: noise removal

Opening followed by closing removes structures and holes smaller than the structuring elements.

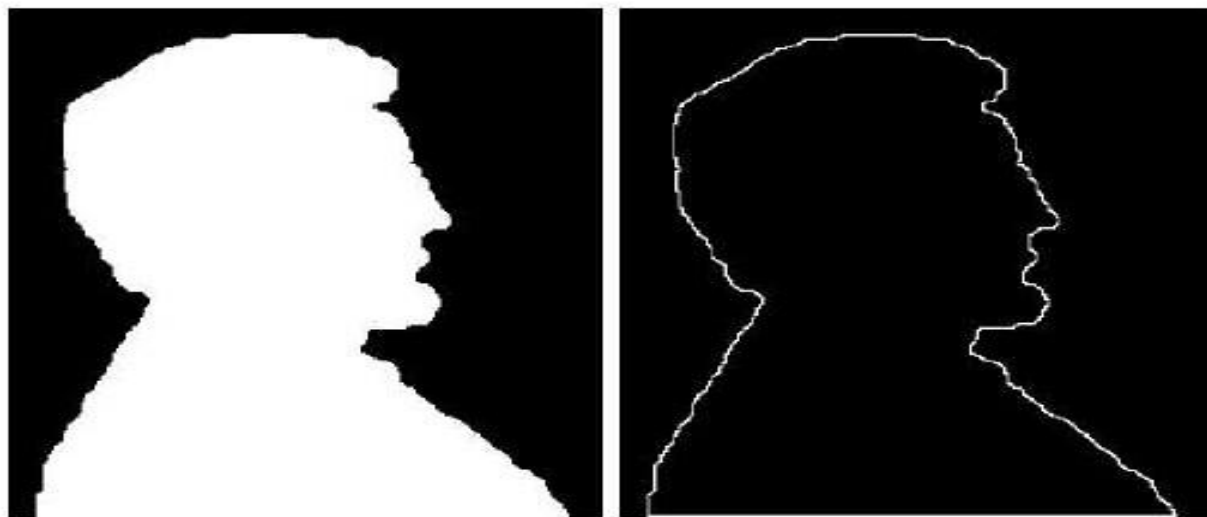


a	b
d	c
e	f

FIGURE 9.11

(a) Noisy image.
 (c) Eroded image.
 (d) Opening of A .
 (d) Dilation of the opening.
 (e) Closing of the opening. (Original image for this example courtesy of the National Institute of Standards and Technology.)

Boundary detection



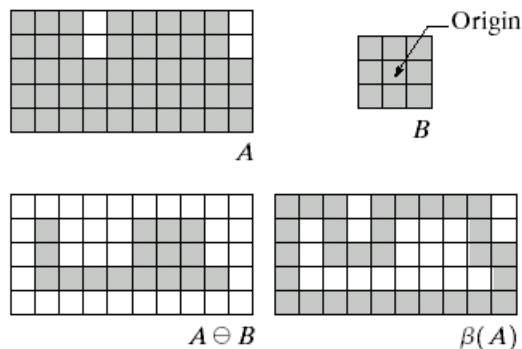
a b

FIGURE 9.14
(a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

$$\beta(A) = A - (A \ominus B)$$

a b
c d

FIGURE 9.13 (a) Set A . (b) Structuring element B . (c) A eroded by B . (d) Boundary, given by the set difference between A and its erosion.



For any query Feel Free to ask

